

THERMAL IMAGING OF ELECTRICAL POWER INSTALLATIONS



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ABSTRACT

This paper provides a set of guidelines for thermal imaging of electrical installations. It begins with a simplified description of the operating principle and areas of application. This is followed by a presentation of the benefits of thermal imaging and the main target group. In the following, critical aspects of the thermal imaging measurement itself and ways to reduce the impact of errors are discussed. An example model for the evaluation of finished thermograms and the anticipated atmosphere for the presentation of the report itself will be considered. In the final part, good practices for preparing for the survey will be described, including the development of the structure of the entire report.

THERMAL IMAGING

Thermal imaging is a non-invasive diagnostic technique that uses a thermal imaging camera to monitor the temperature of an object. The resulting image (called a thermogram) maps the actual heat distribution on the surface using proportional colors. Modern technical solutions allow for high quality measurements. The biggest (and often overlooked) issue is taking into account the influence of environmental variables on measurement accuracy and the correct interpretation of the results obtained. Thermal imaging is becoming increasingly important in a growing technical spectrum.

HOW DOES IT WORK?

Every object with a temperature above absolute zero (above -273.15°C) emits thermal radiation. Everybody has a different emissivity coefficient ϵ . The wavelength of the radiation beam depends on the temperature of the object. The human eye can only perceive a small part of this thermal radiation. The lens of a thermal imaging camera records infrared radiation by measuring energy (thermal signature). The use of thermal imaging cameras is particularly recommended for measuring electrical installations due to their non-invasiveness, thus ensuring the safety of the operator.

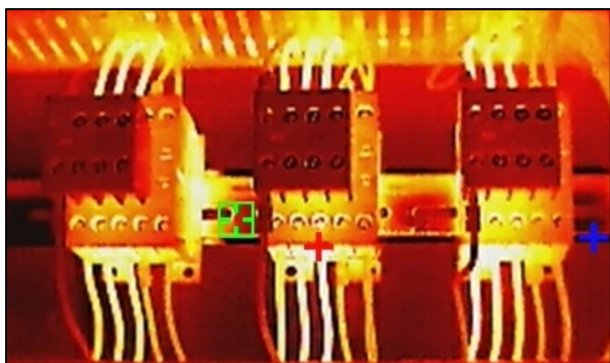


Figure 1. Thermal image: contactors (1)

TEMPERATURE MEASUREMENT

- of power equipment (including switchgear and transformer stations);
- of electronic automation and control systems (including control and measurement equipment);

CONTROL IS AWARENESS

Based on the experience gained, thermal imaging measurement can be part of a strategy to maintain the continuity of equipment operation, such as preventive or predictive maintenance. A rational strategy allows for awareness of the technical condition of equipment and installations, and thus protection against unforeseen events with serious consequences (including fires). Knowing the technical condition of the installation makes it possible to take any necessary action. Regular inspections make it possible to obtain more favourable insurance conditions (including lower costs).



Figure 2. Determine the correct operating temperatures for the devices (2)

BASIC BENEFITS

- control of the technical condition of the installation;
- early detection of damage and taking action;
- prevention of serious failures and downtime;

WHAT THE EYE DOES NOT SEE.. ..MAY COST YOU!

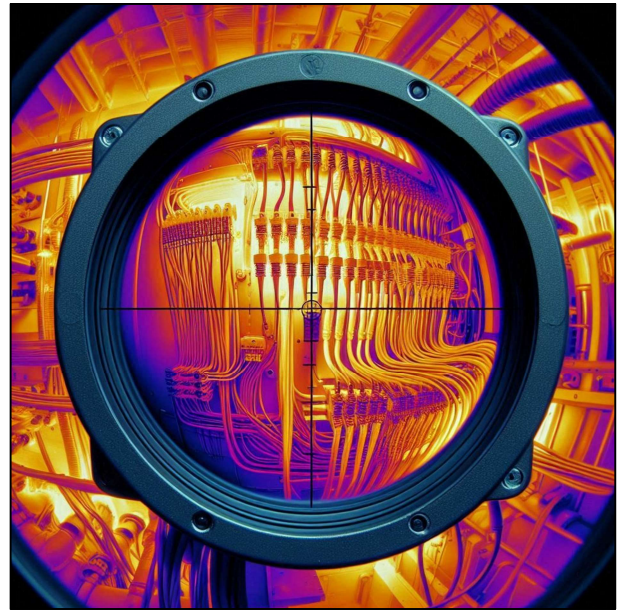


Figure 3. Determine the most stressed components of the installation (3)

Long-term proper operation, visually good technical condition, or the inability to perform invasive electrical measurements (due to required production breaks) are often reasons for neglecting inspections. Thermal imaging allows for safe, non-invasive and comprehensive inspections of electrical installations. Neglecting electrical inspections always has serious consequences. Interruptions in the technological process are one of the least serious consequences of negligence. Wear and tear and aging of equipment eventually lead to failures, resulting in fires and endangering human life.

RECOMMENDED FOR FACILITIES

- critical infrastructure;
- continuous production;
- increased risk of fire;
- outdated electrical installation;
- high insurance costs or requirements;

THE DEVIL IS IN THE DETAILS

The correctness of the thermogram is a combination of factors not only resulting from human error, but also environmental factors, which we try to minimize. For safety reasons, the camera operator performing the electrical installation measurements must have knowledge of electricity (supported by electrical qualifications). The need for electrical knowledge also arises from the need to understand the electrical devices and installations that are the subject of the examination. Novice thermographers often face several negative factors, resulting in a totally abstract thermal imaging measurement.

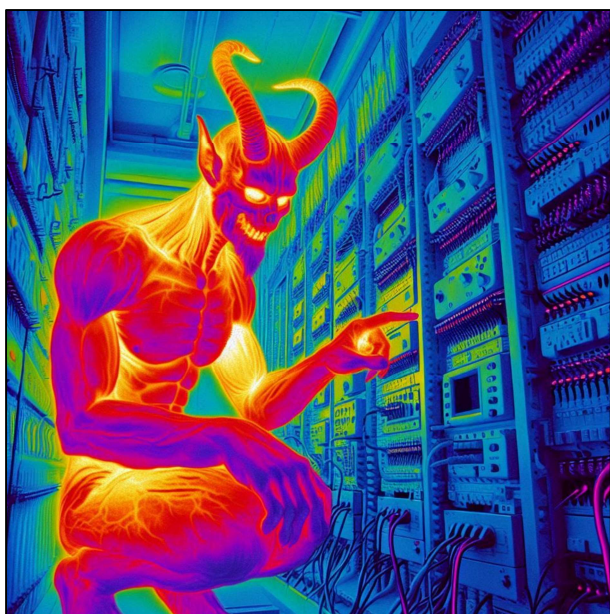


Figure 4. From details to measurement accuracy (4)

TECHNICAL SENSE OF PERFORMING AN EXAMINATION

Visual inspection allows for assessing the technical condition of the installation. Visible damage to devices, rust or water are just some of the basic aspects that speak for refraining from performing measurements. It only makes sense to carry out a thermographic inspection of electrical installations if the installation is under load - ideally at nominal load - reflecting normal and most frequent operating conditions.

The contractor should prepare the installation in advance so that it is neither switched off nor in start-up mode on the day of the inspection.

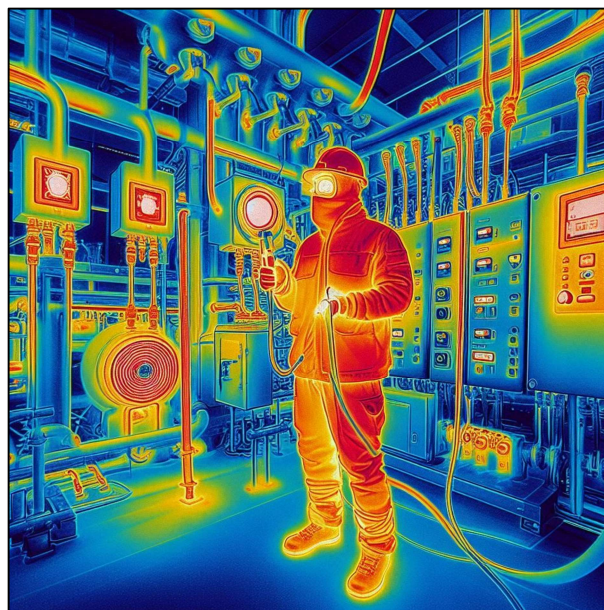


Figure 5. Visual inspection determines the purpose of the tests (5)

BE SURE TO CHECK BEFORE INSPECTION

- the possibility of taking measurements safely;
- the technical condition of the electrical installation;
- the tightness of the housing or installation;
- whether the electrical installation is loaded
- the measurement distance according to the distance set in the camera;
- the possibility of taking measurements directly and perpendicularly;



Figure 6. Take care of your safety (6)

ENVIRONMENTAL FACTORS

We determine the measurement distance by taking a thermal imaging measurement. The distance is a value set in the camera parameters. A value as close as possible to the actual distance increases the accuracy of the measurement. Different angles will generate different readings. The angle at which the measurement is taken must minimize external influences (reflections from light sources). However, the camera lens should be positioned as perpendicular as possible to the object being measured.

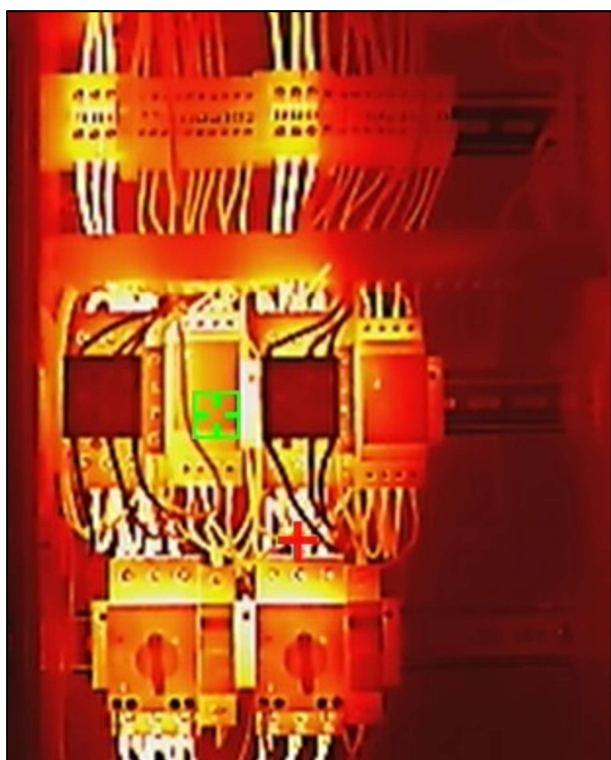


Figure 7. Thermal image: engine control system (7)

TECHNOLOGICAL FACTORS

Environmental factors often prevent correct measurements from being taken in outdoor areas. High humidity and temperatures below the dew point will cause fog to form. Measurements taken in the open air will therefore be more accurate on a dry but cloudy day. It is worth remembering that the greater the temperature difference (at a lower ambient temperature), the more accurate the measurement. It is not always possible to achieve the right ambient conditions. In industrial plants, the air may contain volatile substances: vapours and dust.

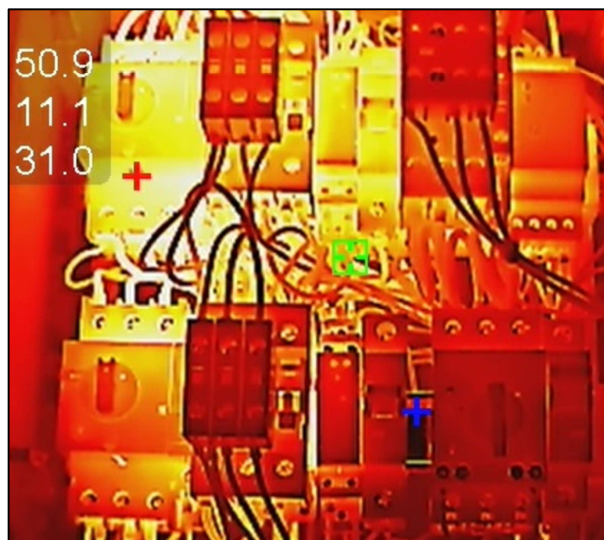


Figure 8. Thermal image: circuit breaker overload (8)

NEGATIVE ENVIRONMENTAL FACTORS

- solar radiation on the measuring object;
- precipitation (rain, snow) and high humidity;
- high ambient temperature;

NEGATIVE TECHNOLOGICAL FACTORS

- vapor and gas;
- solid particles (dust);

SOURCES OF ANOMALIES

It is assumed that the thermographic inspection of electrical installations is direct. The surface of electrical devices or connectors must be accessible. The poor technical condition of the devices not only threatens safety, but also prevents proper measurement. Rust, dirt or dust deposits on the surfaces to be examined completely change the emissivity of these objects - especially when dealing with copper. The same is true when using plastic protective covers. The presence of a transparent cover refracts the wave emitted by the device, but also reflects light from outside. The thermal image taken through the screen will then mostly show the reflection of the camera operator instead of the object being examined. You should also avoid placing a light source in the camera frame.

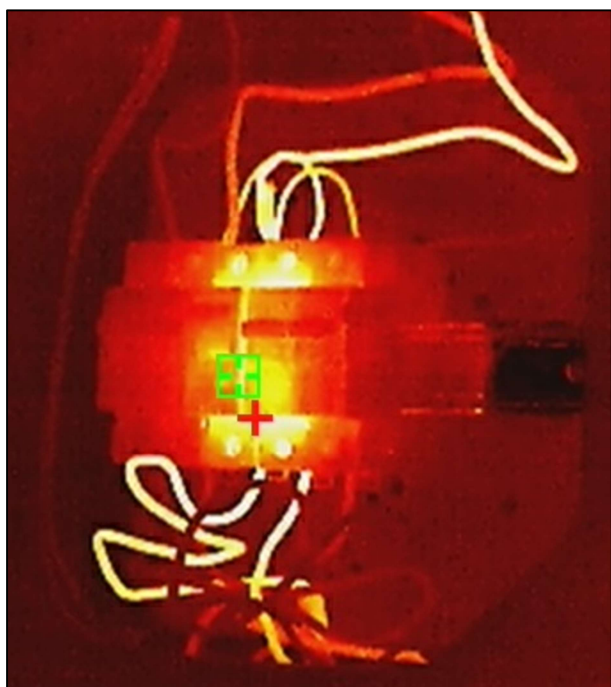


Figure 9. Thermal image: uneven load (9)

FACTORS THAT LEAD TO ANOMALIES

- objects with different emissivity in the camera frame;
- protective objects such as transparent plastic covers;

- objects that reflect light;
- light sources;

SUMMARY OF THE EXAMINATION

We have made correct thermograms of the electrical installations. It is time for the final examination report. With a large amount of data processing, it is worth considering computer support for generating reports, which will calculate and automatically determine the thermal condition of the installation for the entered temperature readings. An app or a simple interface also allows you to add a thermogram to the report. In the future, artificial neural networks may be used to analyse and evaluate thermograms. Each report must have an evaluation note, and in case of irregularities, an additional comment from a certified thermographer.

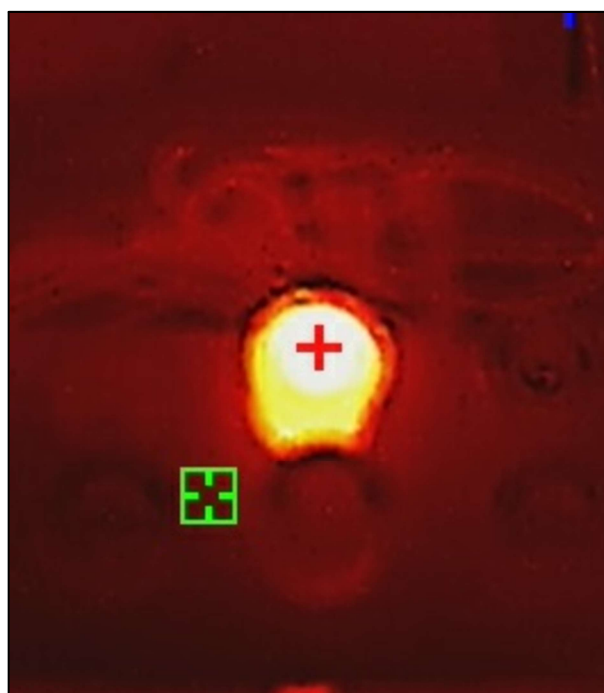


Figure10. Thermal image: overheated fuse (10)

EVALUATION OF THE THERMOGRAM

The thermal evaluation of an electrical installation based on the thermogram is often an individual matter depending on the type of building. The safe temperature range is particularly difficult to determine due to the different thermal resistance of devices and the materials they are made of. The safe temperature range must therefore take into account the thermal resistance of the material (not only short-term but also continuous resistance), its flammability and the age of the installation. It is extremely important to note that the technical condition of the installation is determined by its weakest link. The interpretation of thermograms depends on the temperatures present and requires the development of ranges and scales. The following table (Figure 11) was developed based on VdS and VATH guidelines, which are not standardized due to the individuality of the test objects.

TEMPERATURE OF THE DEVICES IN RELATION TO THE TEMPERATURE OF THE IMMEDIATE SURROUNDINGS	TEMPERATURE OF THE DEVICES IN RELATION TO THE AIR TEMPERATURE	EVALUATION	RECOMMENDED ACTIONS
not exceeding 10°C	not exceeding 65°C	TEMPERATURE OF THE DEVICES NORMAL	no intervention required
not exceeding 10°C	between 65 - 125°C	RECOMMENDED OBSERVATION	observe the device status
between 10 - 25°C	not exceeding 65°C	RECOMMENDED OBSERVATION	
between 10 - 25°C	between 65 - 125°C	INTERVENTION REQUIRED	take countermeasures at the earliest opportunity, observe until then
between 25 - 50°C	not exceeding 65°C	INTERVENTION REQUIRED	
exceeding 50°C	any value	IMMEDIATE INTERVENTION REQUIRED	immediate countermeasures: switch off devices and locate the causes of the status
any value	exceeding 125°C	IMMEDIATE INTERVENTION REQUIRED	

Figure 11. Proposed scale for evaluating thermograms (11)

CORRECT INTERPRETATION, AND THEN.. COMPARISON?

The final report of the thermographic inspection is a document certifying the current technical condition of the installation. Making false or exaggerated assessments in relation to a faulty installation has not only legal consequences, but also endangers human life and health. The client should always expect a negative acceptance of the installation test results - especially those revealing irregularities - after all, he paid for the tests (and therefore assumed their correct result). In order to avoid unpleasantness after the reports have been submitted and at the same time to remain with a clear conscience, it is worth thinking about a compromise beforehand. A compromise for both parties is to introduce a preliminary report with a list of all irregularities and recommendations for remedial action. The client then declares that they will repair the installation within the specified period. After the irregularities have been rectified, a secondary test is carried out on the repaired installation only. The final report of this procedure shows the actual (and correct) condition of the tested installation.

GOOD PREPARATION PRACTICE

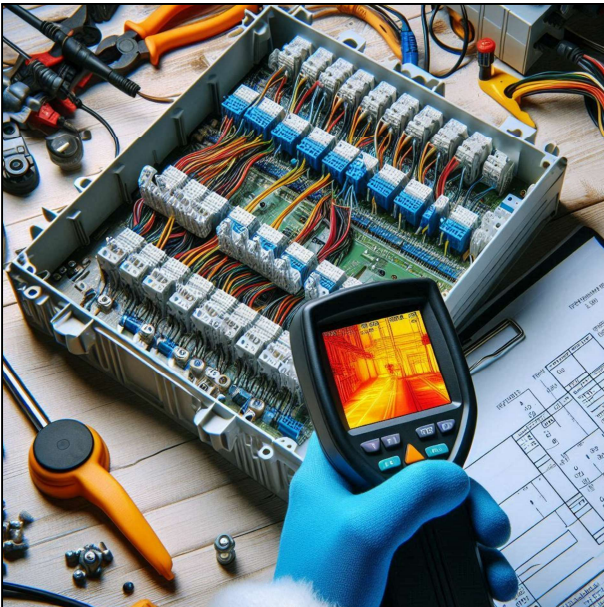


Figure 12. Check the settings of the thermal imaging camera (12)

The basis for documenting the thermal imaging measurements is the thermal imaging report. The report is based on the scope of the inspection as agreed by the client and contractor. It is worth discussing the safe temperature ranges for the equipment at the beginning, taking into account the nature of the equipment's operation and the process technology. When ordering a thermovision inspection of electrical installations, the measuring points should be specified: specific devices or installations. Early cooperation allows for a better understanding and definition of needs.

ORGANIZATION OF THE REPORT

Knowing the scope of work (and often the repetitiveness) makes it possible to prepare for measurements rationally, i.e. to use the measurement point designations precisely. Measurement point designations can be assigned to the location or designation of cabinets, objects or installations. The designations in the report must ensure precise identification of the location of the measurement taken and the specific object being examined. The solution to the problem of multiple independently examined systems is to use introductory pages that open subsequent reports. This solution allows for efficient navigation through the entire report and will be less problematic for the client.



Figure 13. Check the validity of the calibration certificate (13)

PROPOSED STRUCTURE OF THE ENTIRE REPORT

- title page (with the details of the client and contractor);
- introduction and terminology of the terms used (supported by normative acts);
- accepted standard of thermographic evaluation (thermal evaluation of electrical installations);
- scope of measurements (e.g. in the form of an inventory of facilities taking into account the accepted division);
- current calibration certificates of the measuring devices used;
- current certificates and qualifications of thermal imaging camera operators;
- subsequent reports (taking into account the accepted division);
- optionally, a table of contents;

EXAMPLE REPORT: GENERAL DATA SECTION

To keep the report clear, it is a good idea to separate the data and use tables. In the upper section of the document (Figure 14), it is worth including a header containing the date of preparation of the report, the name of the camera operator (and, if available, the service provider's details: company details), the serial number of the thermal imaging camera and other additional measuring devices (if any and if they affect the test result).

<i>Name, address, contact</i>	<i>Logo of the service provider</i>
Report preparation date:	<i>RRRR-MM-DD</i>
Camera model and serial number:	<i>Model and serial number</i>
Other devices(s):	<i>Model and serial number</i>
Object symbol:	<i>Object symbol</i>
Person performing the measurement:	<i>Name and surname</i>
Tested element:	<i>Designation</i>

Figure 14. Sample report organization: upper section (14)

EXAMPLE REPORT: MEASUREMENT AND EVALUATION SECTION

The next section (Figure 15) should contain measurements obtained using additional measuring devices (e.g. thermohygrometer for the measurement environment), followed by measurements from the thermal imaging camera. It is good practice to include information about the settings of the camera itself (e.g. emissivity set, distance of the camera from the object under test). The temperature measurements read from the thermogram (maximum, minimum, average temperature) can be entered into a table. Each report must be concluded with an assessment of the technical condition, including recommendations or comments. It is crucial to include the date and time of the measurements, which precisely determine the technical condition of the installation at a given moment in time.

Category	Value
Date of measurement:	<i>RRRR-MM-DD HH:MM</i>
Measurement taken:	<i>Additional measuring device(s)</i>
<i>Physical quantity A [unit]</i>	<i>Reading A</i>
<i>Physical quantity B [unit]</i>	<i>Reading B</i>
Measurement taken:	Thermal imaging camera
Emissivity coefficient:	<i>Set coefficient value</i>
Set measuring distance: [m]	<i>Set measuring distance</i>
Maximum temperature reading: [°C]	<i>Temperature MAX from thermogram</i>
Minimum temperature reading: [°C]	<i>Temperature MIN from thermogram</i>
Average temperature reading: [°C]	<i>Temperature AVG from thermogram</i>
Additional comment:	<i>Comment</i>
Measurement rating:	<i>Awarded a rating according to the scale</i>

Figure 15. Sample report organization: measurement section (15)

EXAMPLE REPORT: THERMAL IMAGE SECTION

The most important part of the report (Graphic 16) is the actual thermal image. The thermal image is the basis for assessing the technical condition of the equipment - it should therefore be large enough and in focus - so that it is possible to specify each of the tested devices. An additional optical photo of the examined area allows for clarity - especially when the person taking the thermographic photos is not involved in the preparation of the report.

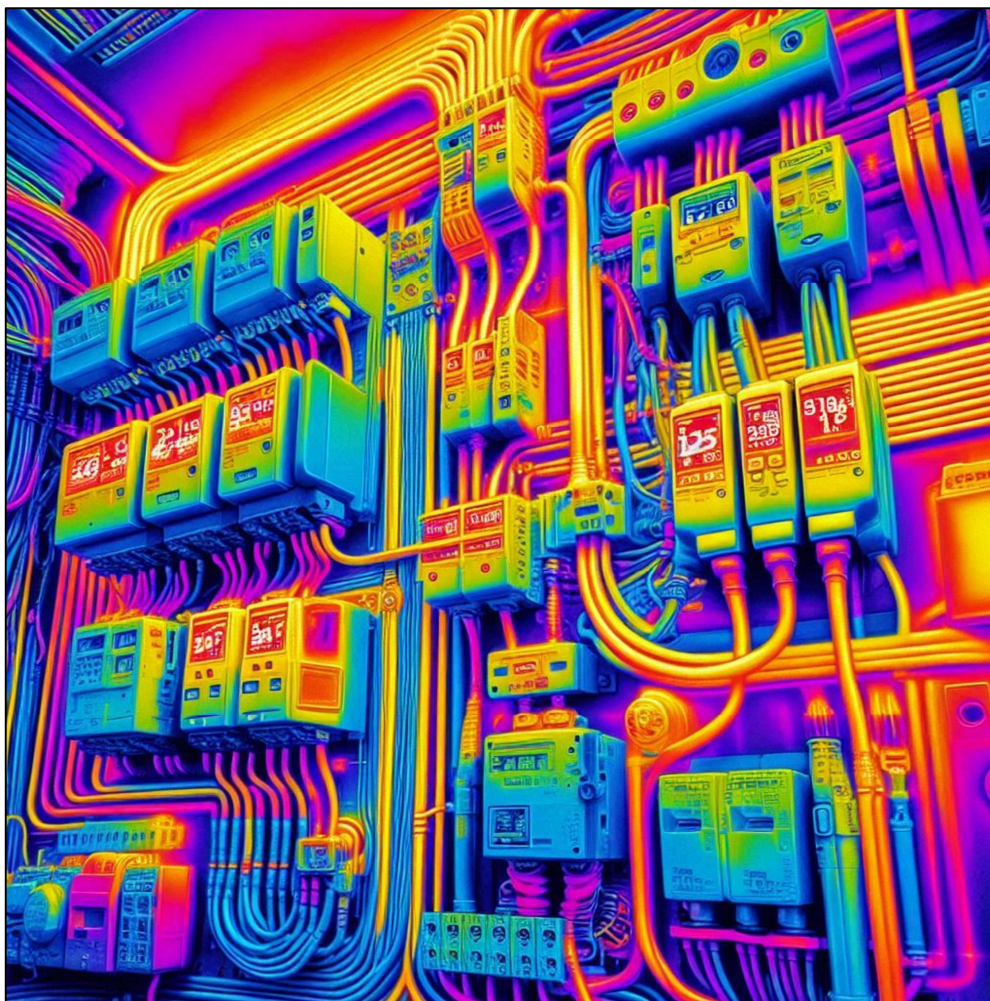


Figure 16. Example report organization: lower section with thermogram (16)

ADDITIONAL COMMENTS

Regardless of the report structure and its visual aspects, it is worth following the principle of simplicity and precision. Justified observations should be described, and remedial recommendations must be based primarily on the safety of the operation. An indisputable confirmation of the authenticity of the measurements taken is an important camera calibration certificate issued by an accredited laboratory. The certificate issued after inspection (and possible calibration) contains the range of permissible deviations in the reference temperature range. Keeping to the inspection deadlines for the calibration of measuring instruments is not only a matter of respecting one's own tool, but also testifies to the high technical culture of the engineer.

CONCLUSION

Thermal imaging is a seemingly simple test when you assume that all you have to do is to place the area to be examined in the camera's field of view and press the shutter release to take the pictures. The tips in this document only give an idea of the problems associated with thermal imaging. The truth is that it is not always possible to eliminate all negative factors, and the survey itself will always be characterized by a certain error. It is particularly important to calibrate the camera, so that not only does our report as a whole gain value, but it also eliminates one of the many errors, which is not difficult at all. I strongly encourage you to try using a thermal imaging camera, as today's cameras are highly intuitive and easy to use. Only by having a thermal imaging camera in your hands and understanding how often anomalies occur in the camera frame will you be able to develop good habits and proper operation.

SELECTED STANDARDS AND GUIDELINES

- ISO 18434-1:2023 - Condition monitoring and diagnostics of machines - Thermography - Part 1: General procedures;
- ISO 18436-7:2014 (E) - Certification of personnel using thermography for machinery diagnostics;
- DIN 54191 - Non-destructive testing - Thermographic testing of electrical installations;
- VDI 2878 - Sheet 1: Application of thermography for maintenance diagnostics:
- Part 1: General requirements and instructions for decision makers and responsible persons,
- Part 2: Electrical installations, Part 3: Equipment technology;
- VdS 2851 - Non-contact temperature measurement (thermography) - practical tips;
- VdS 2858 - Thermography of electrical installations (damage prevention and occupational safety);
- VdS 2859 - Appendix C Minimum requirements for cameras (normative);
- VdS 6021:2022-04 - Thermographic testing of electrical installations;
- PST 6021:2024-01 (03) - Guidelines drawn up by the Association of Thermography Engineers and Technicians;
- Regulation of the Minister of Energy of August 28, 2019, on occupational health and safety with power equipment (consolidated text: Journal of Laws of 2021, item 1210),
- DIN EN 50110-1 (VDE 0105-1) - Operation of electrical installations - Part 1: General requirements,
- VDE 0105 - Operation of electrical installations - Part 100: General requirements

GRAPHICS SOURCES

- Graphics: (2), (3), (4), (5), (6), (12), (13) - generated by artificial intelligence
- Graphics: (14), (15), (16) - as examples based on an original report
- Graphics: (1), (7), (8), (9), (10) - actual, proprietary system measurements
- Table (11) based on a combination of guidelines compiled in PST 6021:2024-01 (03)